

Sale-based Property Valuation Model Development and Analysis

Project 2 | BUSI 444

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Introduction

This report details the development and calibration of a regression model for property valuation. The model is designed to estimate property market values based on the analysis of 466 sales data and relevant property characteristics. The sales data utilized in the model development have been adjusted to reflect market values as of **July 1st, 2017**. The intended users of this valuation model are clients and stakeholders requiring accurate and reliable property value assessments. The scope of this project includes the selection of a suitable valuation model, the classification and transformation of independent variables, and the calibration and validation of the model to ensure accuracy and reliability in its predictive capacity.

This project employs a structured methodology for model development. Initially, a suitable valuation model is selected, and a review of the variables is conducted. Independent variables are then examined to determine their potential applicability to the model. To ensure consistency, sale prices are adjusted to reflect market values as of the valuation date, July 1st, 2017. Following this, the variables are prepared for calibration, and the dataset is divided into training and testing subsets. The model is calibrated using the training dataset and subsequently validated using the testing dataset. The model's final evaluation involves a series of tests to confirm that valuations are fair and unbiased with respect to various property characteristics.

Step 1 – Choose a Valuation Model

The first step in developing the regression model is to define the mathematical structure. The model we will use is designed to estimate the market value (MV) of a property by considering various qualitative and additive factors related to the building, land, and other structures. The general form of the model is as follows:

$$MV = \Sigma GQ_B + \langle \Sigma BQ_B + \Sigma (BQ_M \times BA) \rangle + \langle \Sigma LQ_B + \Sigma (LQ_M \times LA) \rangle + \Sigma (OQ_M \times OA)$$

Where:

- **MV**: Market value of the property.
- **ΣGQ_B** : Sum of general qualitative factors in binary form
- **ΣBQ_B** : Sum of building qualitative factors in binary form
- **BQ_M** : Building qualitative factors in multiplicative form.
- **BA**: Building additive factors.

- ΣLQ_B : Sum of land qualitative factors in binary form
- LQ_M : Land qualitative factors in multiplicative form
- LA : Land additive factors
- OQ_M : Other building qualitative factors in multiplicative form
- OA : Other building additive factors

Following the definition of the valuation model, the subsequent step involves a detailed review of the available independent variables to ensure appropriate categorization for inclusion in the model.

Step 2 – Variable Review

To implement the model, we need to categorize the available independent variables into the appropriate groups. The variables from the dataset can be classified into the following categories:

General Qualitative (GQ variables)

General qualitative variables capture broad locational influences that affect the property's value. These factors include the neighbourhood, proximity to amenities, and other location-specific characteristics. In our dataset, **NEIGH** falls into this category:

Building Quality (BQ variables)

Building quality variables represent the characteristics and condition of the main building. These factors include construction quality, age, and effective year built, which accounts for renovations and depreciation. The following variables are categorized as building quality factors. Since the columns are repeated from Project 1, the data description is not included in the main text, but they are available in the Appendix.

MBMANCLS	MBYRBLT	MBEFFYR
LINMCLS	MBMCLSDV	

Building Additive (BA variables)

Building additive variables include measurable features of the main building, such as floor area, number of rooms, and other physical characteristics. The following variables are categorized as building additive factors:

BEDROOMS	BATHS	FAMILYRM
FIREPLCS	FLRAREA1	FLRAREA2
BSMTFIN	CRWLAREA	BMSTAREA
UNFINBMT	AREA1_2	TOTAREA

BASESTE	HEATTYPE	SPLITVL
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Land Qualitative (LQ variables)

Land qualitative variables represent features of the land that influence its value, such as lot shape, topography, and location-specific characteristics. These variables are often binary, indicating the presence or absence of a feature. The following variables are categorized as land qualitative factors:

CORNER	WOODED	PIE_SHPE
SLOPE	ABV_ROAD	CULDESAC
CONDUIT	ALLEY	SIDEWALK

Land Additive (LA variables)

Land additive variables represent measurable features of the land, such as **LOT_SIZE**

Other Building Qualitative and Additive (OQ and OA variables)

Other building qualitative and additive variables represent features of additional structures on the property, such as garages, carports, and outbuildings. These variables can be qualitative (e.g., quality class) or additive (e.g., area). The following variables are categorized as other building factors:

GARAGEFL	CARPTFL	CP_AREA
GAR_AREA	GAR_TYPE	

Prior to their use in the model, the variables require examination and, where necessary, transformation to ensure their suitability for regression analysis.

Step 3, 4, and 5 – Examine Potential Independent Variables and Transform as Necessary

A critical transformation applied to the data involves adjusting sale prices to a specific date to account for temporal market variations. I need to redo the time adjustment for the sale price based on feedback from the first project.

Time Adjustment for Sale Price

Although the median of SAR (Sale Price Assessed Value Ratio) does not significantly change month over month (project 1), average value of SAR changes month from month and I should ensure the values are adjusted to reflect the value as of July 1st, 2017. The table below shows the variation in the average of SAR in different months. Note that the mean value for SAR is

approximately the SAR in the middle of the month and so the SAR on July 1st is obtained by averaging the mean (SAR) for the month of June and July.

Group	Month	Mean (SAR)	Mean (SAR) on July 1 st	Factor
3	March	1.018	1.031	1.013
4	April	1.032	1.031	0.999
5	May	1.030	1.031	1.001
6	June	1.022	1.031	1.009
7	July	1.041	1.031	0.991
8	August	1.020	1.031	1.011
9	September	1.056	1.031	0.977
10	October	1.053	1.031	0.979

Therefore, the time adjusted sale price will be obtained as follows:

$$\text{Time Adjusted Sale Price} = \text{Factor} \times \text{Sale Price}$$

To ensure that the adjusted sale price is equally distributed over months, I run Kruskal-Wallis test for the adjusted sale price and p-value for the test is 0.186 which is well above the threshold of 5%.

Following what recommended by the textbook, I form the size adjustment factor for land size as follows and calculate the adjusted lot size:

$$\text{Size Factor} = (\text{Average Lot Size} / \text{Lot Size})^{\frac{1.2}{2}}$$

$$\text{Adjusted Lot Size} = \text{Size Factor} \times \text{Lot Size}$$

Doing tries and error, I confirmed that the factor 1.2 works with our data using the size factor obtained for Project 1.

Next the manual classes should be included in the related variables. Manual classes for main building, carport and garage are available but the scaled numeric value is only available for main building. Using the information given in Project 1, I calculated the scaled value for all manual classes (Table A.1)

$$\begin{aligned} \text{Adjusted Floor Area 1} &= \text{Linear MC} \times \text{Floor Area 1} \\ \text{Adjusted Floor Area 2} &= \text{Linear MC} \times \text{Floor Area 2} \\ \text{Adjusted Finished Basement} &= \text{Linear MC} \times \text{Finished Basement} \\ \text{Adjusted Garage Area} &= \text{Linear MC}_{\text{Garage}} \times \text{Garage Area} \\ \text{Adjusted Carport Area} &= \text{Linear MC}_{\text{Carport}} \times \text{Carport Area} \end{aligned}$$



Next the effective year should be calculated by subtracting the effective year built from 2017:

$$\text{Effective Age} = 2017 - \text{Effect Year Built}$$

After the time adjustment, a detailed examination of each variable is needed to analyze its statistical properties, identify potential outliers, and prepare the data for subsequent modeling.

Variable Examination

First, I need to provide a descriptive analysis for each variable to find the range for each and identify outliers if any.

Descriptive Analysis for Continuous Variables

	Min	Max	Mean	Median
flrarea1_adjusted	279.9	2,247.2	1,023.0	1,009.3
flrarea2_adjusted	0.0	1,249.3	32.0	0.0
bsmtfin_adjusted	0.0	1,600.0	439.8	451.5
crwlarea	0.0	1,331.0	57.9	0.0
bsmtarea	0.0	1,978.0	998.9	1,040.0
unfinbmt	0.0	1,826.0	544.6	531.0
area1_2_adjusted	279.9	3,496.5	1,055.0	1,020.6
totarea_adjusted	279.9	3,578.0	1,493.9	1,501.5
lot_size_adjusted	5,131.1	9,635.6	6,998.3	6,891.6
effage	2.0	74.0	25.6	23.0
gar_area_adjusted	0.0	748.8	82.0	0.0
cp_area_adjusted	0.0	663.0	132.8	0.0
saleprice_time_adjusted	68,099.9	305,608.2	148,945.7	144,445.0

According to the table above, there do not appear to be any outliers in the variables. The adjusted floor area for the first-floor ranges from 279.9 to 2,247.2 square feet, which is within a reasonable range. The second-floor area can go up to 1,249.3 square feet, and the adjusted basement finished area reaches up to 1,600 square feet, with some records having zero values in these categories. Crawl space and basement area also show a maximum of 1,331 and 1,987 square feet, respectively, while the unfinished basement area can go up to 1,826 square feet. The adjusted area for both floors combined (area1_2_adjusted) ranges from 279.9 to 3,496.5 square feet, which aligns with the overall distribution of floor and basement areas. Total adjusted area (totarea_adjusted) ranges from 279.9 to 3,578.0 square feet, which again appears reasonable, considering the ranges of individual floor and basement areas. The lot size adjusted variable shows a range of 5,131.1 to 9,635.6 square feet, and this is also within a typical range for residential properties. The effective age of the property (effage) ranges from 2 to 74 years, with a mean of 25.6, suggesting a reasonable mix of newer and older properties in the dataset. Finally, the garage area adjusted variable goes up to 748.8 square feet, with some properties having no garage area (0 value), which is expected. The carport area adjusted (cp_area_adjusted) also shows a range from 0 to 663 square feet, with some zero values. The sale price adjusted (saleprice_time_adjusted) ranges from \$68,099.9 to \$305,608.2, with a mean of 148,945.7, which seems consistent with the property sizes and characteristics observed in the other variables. There do not appear to be any extreme outliers in terms of price, given the variation in property sizes and ages.

Now, I do the same for categorical variable by finding the frequency table.

Frequency Tables

	0	1	2	3	4
bedrooms	0	0	111	335	20
familyrm	164	278	24		
fireplcs	143	215	108		
baseste	442	24			
splitlvl	432	34			
corner	394	72			
wooded	430	36			
pie_shape	439	27			
slope	406	60			
abv_road	385	81			
culdesac	437	29			
conduit	166	300			
alley	318	148			
sidewal	423	43			

garagefl	333	133			
carptfl	268	198			

bath	1	1.5	1.75	2	2.25	2.5	2.75
#	159	113	78	43	41	21	11

heattype	0	1	2	3	4	5	6	7	8
#	0	3	403	14	12	7	12	7	8

The frequency tables above show that there are sufficient records for each category of the categorical variable, except for heat type 1, which contains only three records. Given the description of heat types, it makes sense to merge heat types 1 and 2 as they are both gases. Likewise, heat types 5 and 6 share similar characteristics, so I merged these two groups as well. These adjustments help ensure that each category has an adequate number of records for analysis.

Then, I am examining the relationship between the dependent variable (time adjusted sale price) and independent variables after transformation. Note that I do not include the neighborhood in the modeling as the variable is expected to include many features in the property. I will check the performance of final model in various neighborhood at the end. To start, I check for highly correlated independent variables to prevent multicollinearity.

Correlation between independent variables

	flarea1_adjusted	bsmtfin_adjusted	crwarea	bsmtarea	area1_2_adjusted	totarea_adjusted	conduit	alley	gar_area_adjusted	garagefl	cp_area_adjusted	carptfl
flarea1_adjusted	1.00	0.40	-0.24	0.72	0.89	0.80	0.43	-0.42	0.19	0.24	0.16	0.08
bsmtfin_adjusted		1.00	-0.34	0.47	0.26	0.82	0.25	-0.27	0.08	0.06	0.25	0.20
crwarea			1.00	-0.82	-0.17	-0.33	-0.30	0.30	0.03	0.04	-0.11	-0.07
bsmtarea				1.00	0.60	0.67	0.39	-0.38	0.10	0.11	0.17	0.09
area1_2_adjusted					1.00	0.77	0.35	-0.34	0.25	0.28	0.11	0.06
totarea_adjusted						1.00	0.37	-0.38	0.20	0.21	0.23	0.17
conduit							1.00	-0.81	-0.01	-0.07	0.16	0.13
alley								1.00	0.04	0.10	-0.19	-0.15
gar_area_adjusted									1.00	0.77	-0.26	-0.26
garagefl										1.00	-0.36	-0.38
cp_area_adjusted											1.00	0.89
carptfl												1.00

The table shows that the first-floor area is highly correlated with the basement area, total area, and the combined first and second-floor area, which is expected. The finished basement area is also correlated with the total area. Since the total area includes the first and second floors as well as the basement, this overlap makes sense. Crawl space and basement areas are also correlated. Similarly, alley and conduit show a strong correlation. Additionally, garage area and garage presence, as well as carport area and carport presence are all highly correlated.

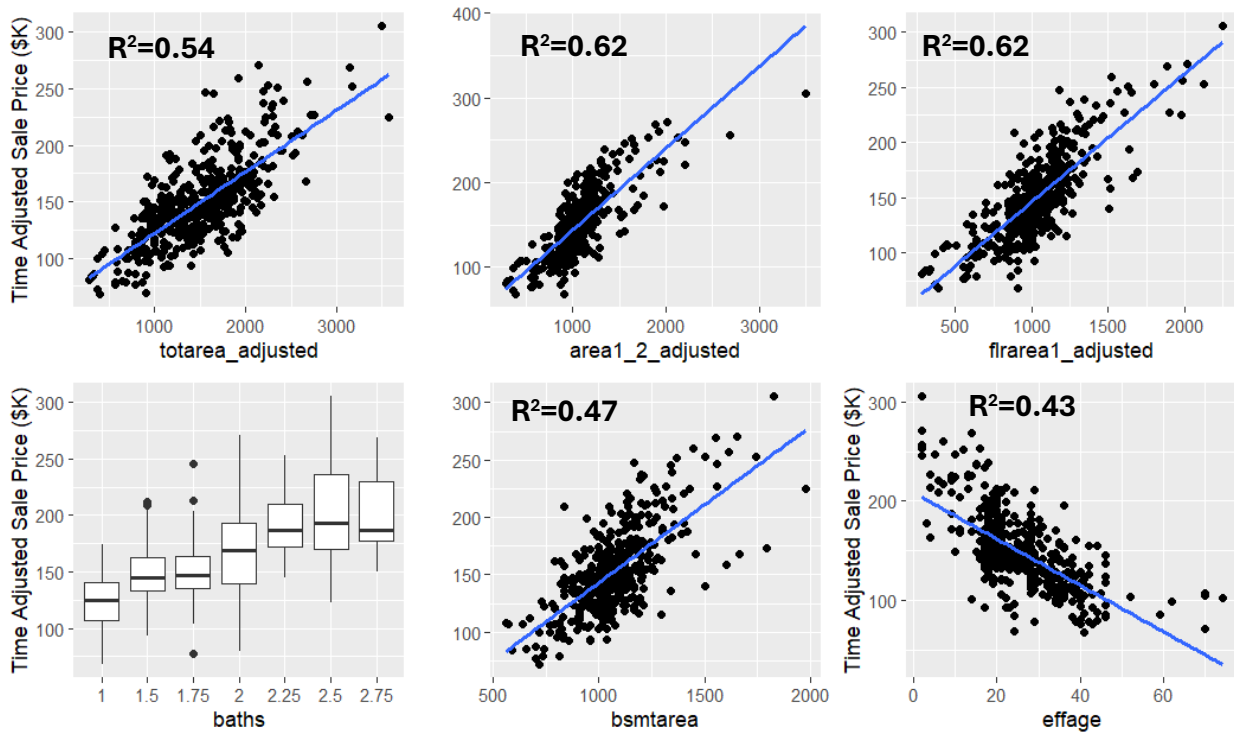
To avoid multicollinearity, only one variable from each group of highly correlated variables should be included in the model. The decision on which variable to keep depends on its correlation with the time-adjusted sale price. The variable with the higher correlation should be kept, and the one with a lower correlation should be excluded. The correlation between the time-adjusted sale price and the independent variables is shown in the table below. Variables highlighted in green have a correlation of 0.7 or above, those highlighted in yellow have a correlation between 0.2 and 0.6, and the remaining variables are not highlighted.

Correlation of Independent Variables with Response Variables

totarea_adjusted	0.738	familyrm	0.386	culdesac	0.299
area1_2_adjusted	0.785	bsmtfin_adjusted	0.407	splitlvl	0.279
flrarea1_adjusted	0.791	fireplcs	0.515	cp_area_adjusted	0.187
baths	0.651	conduit	0.566	baseste	-0.068
bsmtarea	0.626	unfinbmt	0.225	abv_road	-0.054
effage	-0.658	pie_shpe	0.242	slope	-0.033
alley	-0.499	gar_area_adjusted	0.250	sidewalk	-0.112
crwllarea	-0.304	flrarea2_adjusted	0.256	heatttype	0.104
bedrooms	0.364	lot_size_adjusted	0.368	carptfl	0.118

Before selecting variables, it is important to check the plots of the independent variables against the time-adjusted sales. I have included plots for the highly correlated variables (highlighted in green) here. The plots for the other variables are available in the Appendix (Fig A.1 – A.4). For continuous variables, scatter plots with R-squared values are included, while for categorical variables, box plots are used.

Note that in the scatter plots, independent variables with zero value are excluded. For example, the plot for basement area only includes those folios with basement area more than zero. Based on the tables and figures above, I selected the following potential variables for the model. Variables with low correlation to the response variable (<0.1) or low R^2 were excluded, as they don't appear relevant.



Variables to be included and excluded in the model.

Include			Exclude
familyrm	flrarea2_adjusted	sidewalk	unfinbmt
fireplcs	baseste	totarea_adjusted	corner
flrarea1_adjusted	Heattype	bsmtarea	wooded
bsmtfin_adjusted	splitlvl	totarea_adjusted	slope
crwldata	lot_size_adjusted	alley	abv_road
area1_2_adjusted	pie_shpe	Bedroom	gar_area_adjusted
conduit	culdesac	Bath	cp_area_adjusted
garagefl	effage		carptfl

Note that, however, the following variables have high correlation with each other and only one from each group should be included in the modeling.

Group				
1	flrarea1_adjusted	bsmtarea	area1_2_adjusted	totarea_adjusted
2	bsmtfin_adjusted	totarea_adjusted		
3	crwldata	bsmtarea		
4	conduit	alley		

The preceding steps provide the necessary data preparation and relationship identification. The next step involves the selection of specific variables for model calibration.

Step 6 – List Variables for Calibration

Results of the previous steps show that the variables with high correlation with time adjusted sale price and those with high correlation with each other. Following the textbook, I am classifying the variables into various groups and fit the model on each to find the best groups of variables. Among those, there are variables that are common between the groups.

familyrm	flrarea2_adjusted	lot_size_adjusted
fireplcs	baseste	pie_shpe
Heattype	effage	culdesac
splitlvl	sidewalk	garagefl
Bedroom	Bath	

Besides these common variables, I form the following groups as well, each customized based on the variables with high correlation.

Groups of Variables

Group 1 (19 Variables)	flrarea1_adjusted	bsmtfin_adjusted	crwlarea	alley
Group 2 (18 Variables)	bsmtarea	bsmtfin_adjusted	conduit	
Group 3 (18 Variables)	totarea_adjusted	crwlarea	conduit	
Group 4 (19 Variables)	area1_2_adjusted	bsmtfin_adjusted	crwlarea	conduit

I fitted four models using the variables from the groups above and compared them in the following table. Based on the results, models with Group 4 variables have the best Adjusted R^2 and the lowest SSE. I need to check for collinearity among the variables. Rows highlighted in orange indicate variables with VIF values greater than 3.33, suggesting potential multicollinearity.

Group Number	Adjusted_R2	SSE	F_Statistic	Significance
1	0.79	287283850	96.58	0.00
2	0.77	304138868	95.08	0.00
3	0.78	296209197	98.33	0.00
4	0.80	274373044	102.29	0.00

Group 4

Variable	Coefficient	Standard_Coefficient	t_Statistic	Significance	VIF
(Intercept)	30589.436	0.832	2.660	0.008	
familyrm	-3713.279	-0.101	-1.877	0.061	2.081
flrarea2_adjusted	-9.036	0.000	-1.132	0.258	2.081
lot_size_adjusted	6.010	0.000	3.966	0.000	1.331
fireplcs	3060.934	0.083	2.309	0.021	1.591
baseste	-4483.907	-0.122	-1.138	0.256	1.288
pie_shpe	5873.225	0.160	1.533	0.126	1.360
heattype	-75.964	-0.002	-0.122	0.903	1.059
effage	-710.294	-0.019	-5.653	0.000	2.875
culdesac	4699.267	0.128	1.263	0.207	1.371
splitlvl	6209.205	0.169	1.967	0.050	1.145
sidewalk	3181.343	0.087	1.128	0.260	1.131
garagefl	8855.042	0.241	4.716	0.000	1.221
area1_2_adjusted	50.936	0.001	9.744	0.000	4.121
bsmtfin_adjusted	11.051	0.000	2.917	0.004	2.644
crwlarea	-11.454	0.000	-3.005	0.003	1.245
conduit	11682.323	0.318	4.804	0.000	2.303
bedrooms	3527.746	0.096	1.905	0.057	1.416
baths	10064.459	0.274	4.415	0.000	2.179

The selected model presents a few issues. area1_2_adjusted shows VIF values greater than 3.33, indicating potential multicollinearity with other variables. This is expected since area1_2_adjusted are correlated with effage, leading to a high VIF. However, due to the importance of area1_2_adjusted, I keep it for now and will revisit its inclusion during model calibration.

Lastly, I am analyzing the coefficients to ensure they make sense. familyrm has a negative coefficient, which may not align with expectations. However, after examining the large positive coefficient for bedroom, it appears that houses with more bedrooms are preferred over those with larger family rooms, as more bedrooms make the house more functional. The negative coefficient for flrarea2_adjusted is also unexpected, but since its effect is not statistically significant, we can leave it as is and expect this issue to be addressed during the stepwise approach. Similarly, the variable baseste, which indicates the presence of a basement, shows a negative coefficient, but its effect is also not significant and will be reviewed further in the stepwise analysis. To facilitate model training and validation, the dataset was partitioned into training and testing subsets. The methodology for this partitioning is described in the following section.

TEST/MODEL Split

Our selected model has 26 variables, requiring at least 130 sales to calibrate the parameters. This is significantly lower than the number of sales available in our dataset. I allocated 320 sales for model training and reserved the remaining 146 for testing (a 70/30 split). The original dataset includes a random column for sorting sales. I used this column to sort the data and selected the first 320 sales for training (Step 7) and the last 146 for testing (Step 8).

With the training dataset established, the model parameters require calibration. The following section details the stepwise approach utilized for this calibration.

Step 7 – Model Calibration

I use stepwise approach to calibrate the parameters; the model's summary for each step is included in Appendix (Table A.2) and the model summary with the coefficients are included here (Table A.3).

Final Model ANOVA table

	SSE	df	Mean SSE	F	Sig
Regression	2.93659E+11	14	20975655466	74.04979	0.00
Residuals	86395588388	305	283264224.2		
Total	3.80055E+11	319			

Final Model Coefficient

Variable	Coefficient	Standard_Coefficient	t_Statistic	Significance	VIF
(Intercept)	35582.070	1.031	2.605	0.010	
familyrm	-4334.855	-0.126	-1.751	0.081	2.181
lot_size_adjusted	5.032	0.000	2.642	0.009	1.314
fireplcs	3378.705	0.098	2.135	0.034	1.477
baseste	-8409.833	-0.244	-1.620	0.106	1.274
pie_shpe	10423.479	0.302	2.265	0.024	1.203
effage	-536.207	-0.016	-3.485	0.001	2.451
splitlvl	6654.107	0.193	1.627	0.105	1.158
sidewalk	6965.045	0.202	1.865	0.063	1.093
garagefl	6417.647	0.186	2.765	0.006	1.188
area1_2_adjusted	56.863	0.002	11.162	0.000	2.150
bsmtfin_adjusted	15.497	0.000	3.329	0.001	2.488
crlarea	-12.359	0.000	-2.536	0.012	1.253
conduit	13379.662	0.388	4.742	0.000	2.054
baths	8769.237	0.254	3.120	0.002	2.017

Excluded Variable

Variable	Coefficient	Standard_Coefficient	t_Statistic	Significance	VIF
flrarea2_adjusted	-9.036	0.000	-1.132	0.258	2.081
heattype	-75.964	-0.002	-0.122	0.903	1.059
culdesac	4699.267	0.128	1.263	0.207	1.371
bedrooms	3527.746	0.096	1.905	0.057	1.416

As expected, all the variables that were excluded have t-statistics with significance levels of 21% or higher, meaning they are not statistically significant at the 10% level. Similarly, most of the variables included in the final model after the stepwise selection are significant at the 10% level. However, variables such as baseste and sidewalk are included in the model, even though their t-statistics are slightly below the 10% significance threshold. On the positive side, none of the variables have a VIF greater than 2.5, indicating that the multicollinearity issues discussed in the previous is not the case anymore.

Next, I am performing a Casewise diagnostic test to check for records with standardized residuals greater than 3 or less than -3. No records meeting this criterion were found, indicating that there are no extreme residuals in the model. Table below shows the normalized value for residual and predicted values.

Residuals and Predicted Values

Metric	Min	Max	Mean	Std. Deviation	Count
Predicted Values	65924.12	272930.60	147799.40	30340.76	320
Residuals	-42981.94	49403.95	0.00	16456.99	320
Std. Predicted Values	-2.70	4.12	0.00	1.00	320
Std. Residuals	-2.64	2.99	0.00	1.00	320

Model Summary

R	R ²	Adjusted R ²	Std. Error of Estimate
0.88	0.77	0.76	16830.46

The negative coefficient for familyrm has already been discussed. Here, I will finalize this step by addressing the variables baseste and crwlarea. Both the presence of a basement (baseste) and crawl areas (crwlarea) came with negative coefficients. It is important to note that the finished basement area (bsmtfin_adjusted) has a positive coefficient, which reflects

the desirability of a finished basement. However, the basement may now be correlated with unfinished areas that are less desirable. Similarly, crawl spaces are generally not as desirable because they are less practical compared to finished basements. Therefore, it is not surprising that the effect of crawl spaces is predicted to be negative in the model.

The final model, therefore, is given below:

saleprice_time_adjusted = 35582.070 - 4334.855 (familyrm) + 5.032 (lot_size_adjusted) + 3378.705 (fireplcs) - 8409.833 (baseste) + 10423.479 (pie_shpe) - 536.207 (effage) + 6654.107 (splittvl) + 6965.045 (sidewalk) + 6417.647 (garagefl) + 56.863 (area1_2_adjusted) + 15.497 (bsmtfin_adjusted) - 12.359 (crlwarea) + 13379.662 (conduit) + 8769.237 (baths)

Ratio Statistics for ASR = Predicted/Time Adjusted Sale Price – Training Data

Mean		1.01
95% Confidence Interval for Mean	Lower Bound	1.00
	Upper Bound	1.03
Median		1.00
95% Confidence Interval for Median	Lower Bound	0.99
	Upper Bound	1.02
	Actual Coverage	95.6%
Minimum		0.75
Maximum		1.52
Coefficient of Dispersion		8.9%

The final phase of model development involves a rigorous assessment of the model's predictive performance using the holdout sample. The results of this testing are presented in this section.

Step 8 – Model Testing

Neighborhood Adjustment

As mentioned earlier, I did not include the neighbourhood in the modeling process. Therefore, it is necessary to test if all the neighbourhoods are valued equally. I am using Kruskal-Wallis test and ratio statistics analysis for this part.

Ranks of ASR for Neighborhoods

Neigh	N	Mean Rank
713	55	155
715	45	226
716	111	151
731	20	34.2

732	5	139
734	10	307
736	18	206
737	56	139

Test Statistics for ASR - Neighborhood

Kruskal-Wallis H	93.95
df	7
Asymp Sig.	0.000

The tables above show that the adjustment for neighborhood is required as the p-value for Kruskal Wallis test is below out 5% threshold. To find out the adjustment ratio, I am checking the ratio statistics for ASR.

Ratio Statistics for ASR

Neigh	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
713	0.99	0.96	1.03	95.6	0.78	1.32	8.56
715	1.08	1.05	1.13	94.6	0.86	1.38	7.45
716	1	0.99	1.02	95.3	0.82	1.22	5.95
731	0.83	0.78	0.87	92.2	0.75	0.98	7.59
734	1.31	1.2	1.35	93.5	1.11	1.52	7.16
736	1.09	0.97	1.1	93.6	0.89	1.24	6.65
737	0.99	0.96	1.01	93.7	0.76	1.22	7.94

Table above shows that neighbourhood 715, 731, and 734 need adjustment as the confidence interval for median does not include 1.00. I adjust the prediction for neighbourhood 715 by $1/1.08$, neighborhood 731 by $1/0.83$, and neighborhood 734 by $1/1.31$. I repeat the ratio statistics after adjustment in the neighborhood to ensure adjustment is done correctly. The ratio statistics after adjustment is included in the Appendix (Table A.4). Similarly, I am doing more adjustment based on Manual Class, and the ratio statistics before and after the adjustment are available in the Appendix (Table A.5 – A.6). Finally, I check for bedrooms, heattype, culdesac, and alley as I removed them during the stepwise process and their Kruskal-Wallis statistics are all above 5% threshold which shows no further adjustment is needed (Table A.7).

Evaluating Model Quality Using Holdout Sample

In this section, I am testing the model obtained in the previous section using the TEST data. First, let's recreate the ratio statistics for the testing data. The ratio statistics show that the mean and median are very close to 1.00 and the confidence interval for both include 1.00. The COD is also within the IAAO standard of 5 to 10%, which is acceptable.

Ratio Statistics for ASR = Predicted/Time Adjusted Sale Price – Testing Data

Mean		1.00
95% Confidence Interval for Mean	Lower Bound	0.98
	Upper Bound	1.02
Median		1.01
95% Confidence Interval for Median	Lower Bound	0.99
	Upper Bound	1.02
	Actual Coverage	95.3%
Minimum		0.67
Maximum		1.34
Coefficient of Dispersion		8.5%

Evaluating Model Quality Using Entire Dataset

Ratio Statistics for ASR = Predicted/Time Adjusted Sale Price – Entire Dataset

Mean		1.01
95% Confidence Interval for Mean	Lower Bound	1.00
	Upper Bound	1.02
Median		1.00
95% Confidence Interval for Median	Lower Bound	1.00
	Upper Bound	1.01
	Actual Coverage	94.82%
Minimum		0.72
Maximum		1.4
Coefficient of Dispersion		7.6%

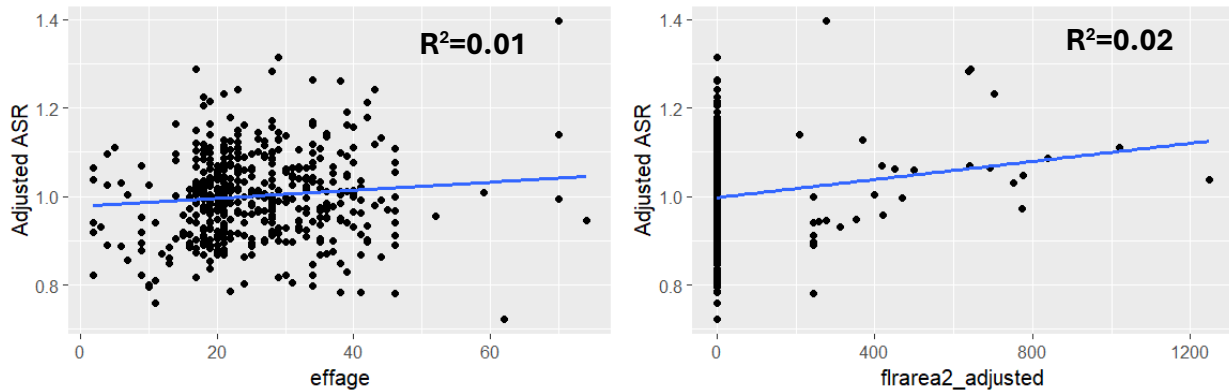
Again, the ratio statistics for entire dataset shows that the COD and median ASR is within IAAO threshold.

Modeling ASR

Next process for testing the model is to ensure that the adjusted ASR does not have high correlation with any of the independent variables I used in the model. That is to show there is no bias for ASR with respect to the independent variables. For that purpose, I fit a model with adjusted ASR as the response variable and the same independent variable as the model above. The model's coefficients are given below.

Variable	Coefficient	Standard_Coefficient	t_Statistic	Significance	VIF
(Intercept)	0.987	10.016	14.924	0.000	
familyrm	-0.010	-0.101	-0.874	0.382	2.081
flrarea2_adjusted	0.000	0.001	2.616	0.009	2.081
lot_size_adjusted	0.000	0.000	-0.973	0.331	1.331
fireplcs	-0.007	-0.067	-0.870	0.385	1.591
baseste	-0.035	-0.352	-1.530	0.127	1.288
pie_shpe	0.033	0.334	1.494	0.136	1.360
heatttype	-0.002	-0.017	-0.464	0.643	1.059
effage	0.003	0.027	3.646	0.000	2.875
culdesac	-0.022	-0.218	-1.007	0.315	1.371
splitlvl	0.014	0.145	0.789	0.431	1.145
sidewalk	0.056	0.568	3.450	0.001	1.131
garagefl	-0.005	-0.049	-0.450	0.653	1.221
area1_2_adjusted	0.000	0.000	0.705	0.481	4.121
bsmtfin_adjusted	0.000	0.000	0.415	0.679	2.644
crlwarea	0.000	0.000	-0.288	0.773	1.245
conduit	0.034	0.349	2.459	0.014	2.303
bedrooms	-0.018	-0.188	-1.736	0.083	1.416
baths	0.016	0.164	1.229	0.220	2.179

There are 4 variables where significantly affect the adjusted ASR and so need to investigate further. Similar to the previous section, I first check if adjustment is needed for conduit and sidewalk in the Appendix. Results show that adjustment for sidewalk is needed which I include the results before and after adjustment in the Appendix (Table A.8 – A.9). Next, I need to check the adjusted ASR with the effage and flrarea2_adjusted to see what is the R^2 . The figures below show that the correlation between the adjusted ASR and effage or floor area 2 is not significant.



Final Check for Neighborhood

As the final check, I do a Kruskal-Wallis test to ensure the adjusted ASR is equally valued across neighborhood. Table below shows that the significance level of the test is not below 5%, our threshold.

Test Statistics for Adjusted ASR - Neighborhood

Kruskal-Wallis H	13.8
df	7
Asymp Sig.	0.05

Conclusion

The final regression model developed in this analysis demonstrates a reasonable capacity for estimating property values. The model's performance is supported by the ratio study and bias tests, which indicate that the model meets IAAO standards. Specifically, the coefficient of dispersion (COD) for the entire dataset is 7.6%, falling within the acceptable range of 5% to 10%, and the median assessment-to-sales ratio (ASR) is 1.00, demonstrating minimal bias. The model also performs acceptably on the holdout sample, with a COD of 8.5% and mean and median ASR values close to 1.00.

Despite its utility, the cost-based model has limitations. Its accuracy is contingent on the quality and completeness of the input data, and it assumes that past sales reliably predict future market values, which may not always be true. While the model incorporates key property characteristics, it may not capture all factors influencing property values. The model's strengths include its objectivity, consistency, and the use of stepwise regression to minimize multicollinearity and identify significant variables.

To enhance the model, several recommendations are made. These include improving data quality control, continuous market monitoring and model recalibration, exploring the

inclusion of additional relevant variables, refining neighborhood-specific adjustments, and ongoing model validation. Addressing these limitations and implementing these recommendations will further strengthen the model's effectiveness as a valuation tool.

Appendix

A.1. Data Description

Building Quality (BQ variables)

- **MBMANCLS:** Indicates the main building class.
- **MBYRBLT:** The year the main building was constructed.
- **MBEFFYR:** The effective year built, considering renovations and depreciation.
- **LINMCLS:** A scaled version of the main building class
- **MBMCLSDV:** The percentage deviation from the main building quality class

Building Additive (BA variables)

- **BEDROOMS:** Number of bedrooms in the property.
- **BATHS:** Number of bathrooms.
- **FAMILYRM:** Count of family or recreation rooms.
- **FIREPLCS:** Number of fireplaces.
- **FLRAREA1:** Area of the first floor.
- **FLRAREA2:** Area of the second floor.
- **BSMTFIN:** Finished basement area.
- **CRWLAREA:** Crawl space area.
- **BMSTAREA:** Full-height basement area.

- **UNFINBMT:** Unfinished basement area.
- **AREA1_2:** Total area of the first and second floors.
- **TOTAREA:** Total finished area, including the basement.
- **BASESTE:** Binary – indicates if there is a basement suite.
- **DECKAREACOVERED:** Covered deck area.
- **DECKAREAUNCOVERED:** Uncovered deck area.
- **HEATTYPE:** The type of heating system, which can influence the property's desirability and value.

Land Qualitative (LQ variables)

- **CORNER:** Indicates if the lot is a corner lot.
- **WOODED:** Indicates if the lot is more wooded than typical lots in the neighbourhood.
- **PIE_SHPE:** Indicates if the lot is pie shaped.
- **SLOPE:** Indicates if the lot is sloped.
- **ABV_ROAD:** Indicates if the lot is above the road level.
- **CULDESAC:** Indicates if the lot fronts on a cul-de-sac.

Other Building Qualitative and Additive (OQ and OA variables)

- **GARAGEFL:** Indicates the presence of a garage.
- **CARPTFL:** Indicates the presence of a carport.
- **CP_AREA:** Total area of the carport.
- **GAR_AREA:** Total area of the garage.

A.2. Variable Transformation

Table A 1. Mapping between Manual Class and Scaled Numeric Value

Garage/Carport Manual Class (factors)	Scaled Numeric Value
910/911	0.450
920/921	1.000
930/931	1.200
940/941	1.420

Variable Examination

Fig A 1. Variable Correlation with Response Variable – Moderate Correlation

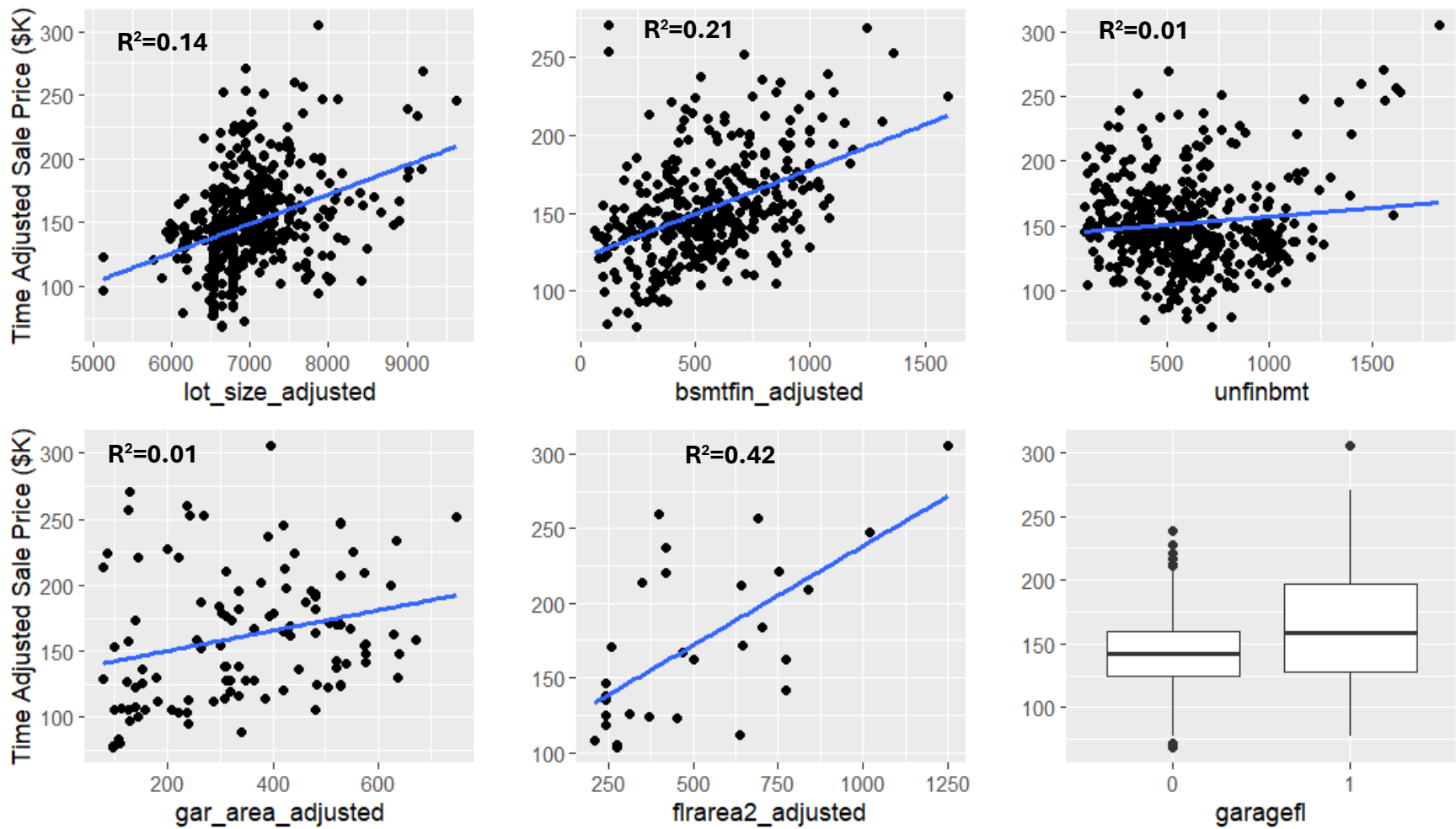
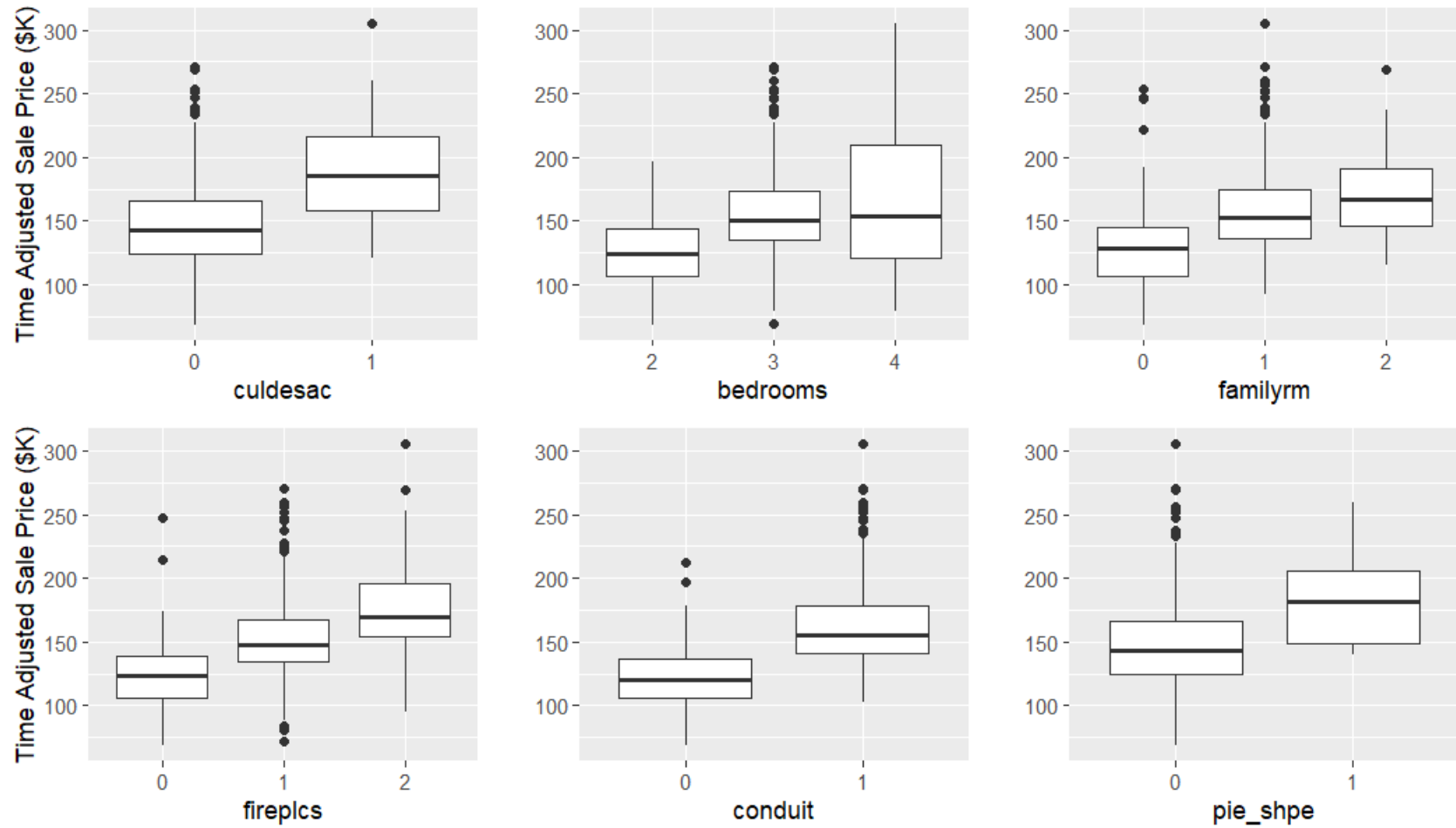


Fig A 2. Variable Correlation with Response Variable – Moderate Correlation



R²=0.33

Fig A 3 Variable Correlation with Response Variable – Low Correlation

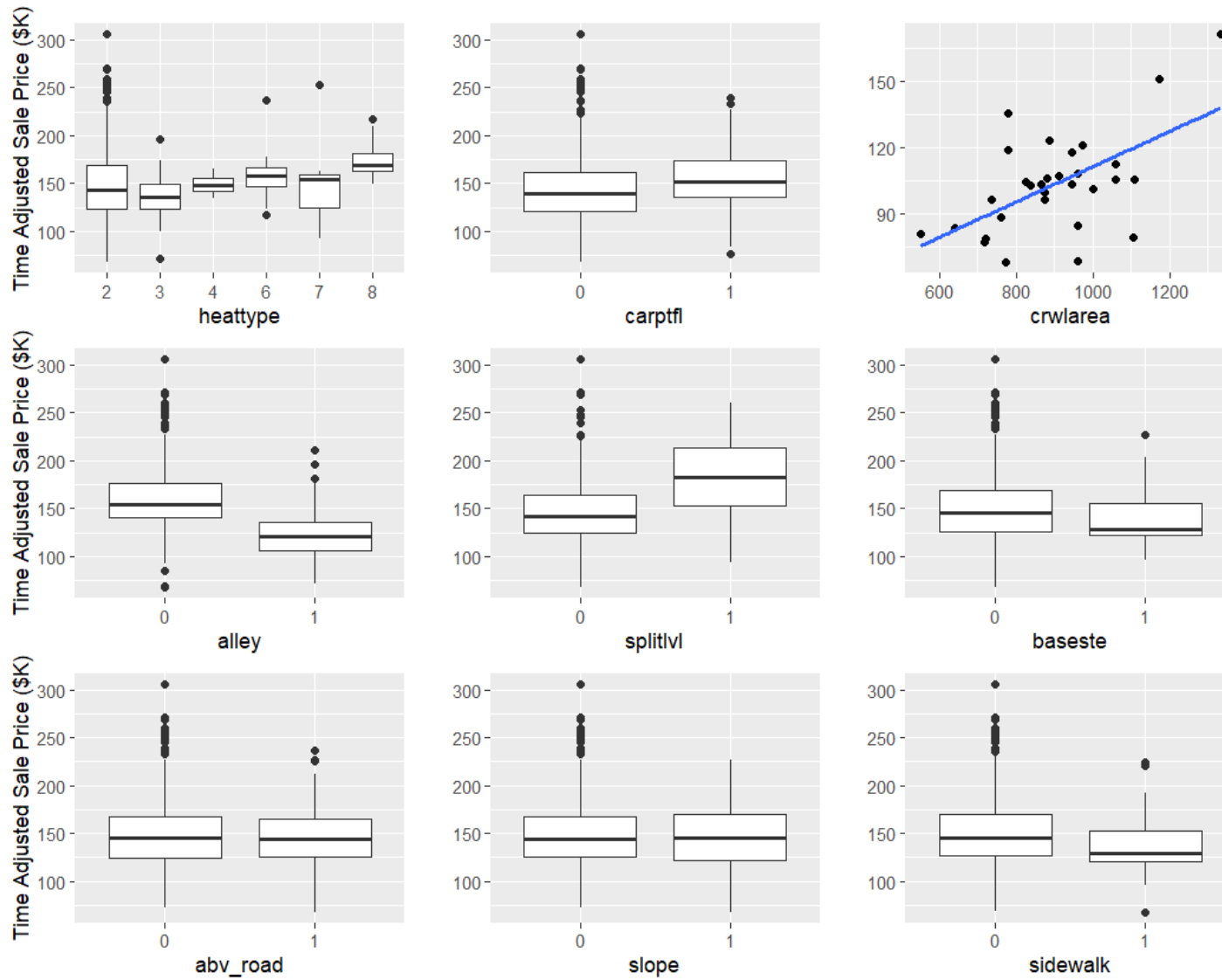
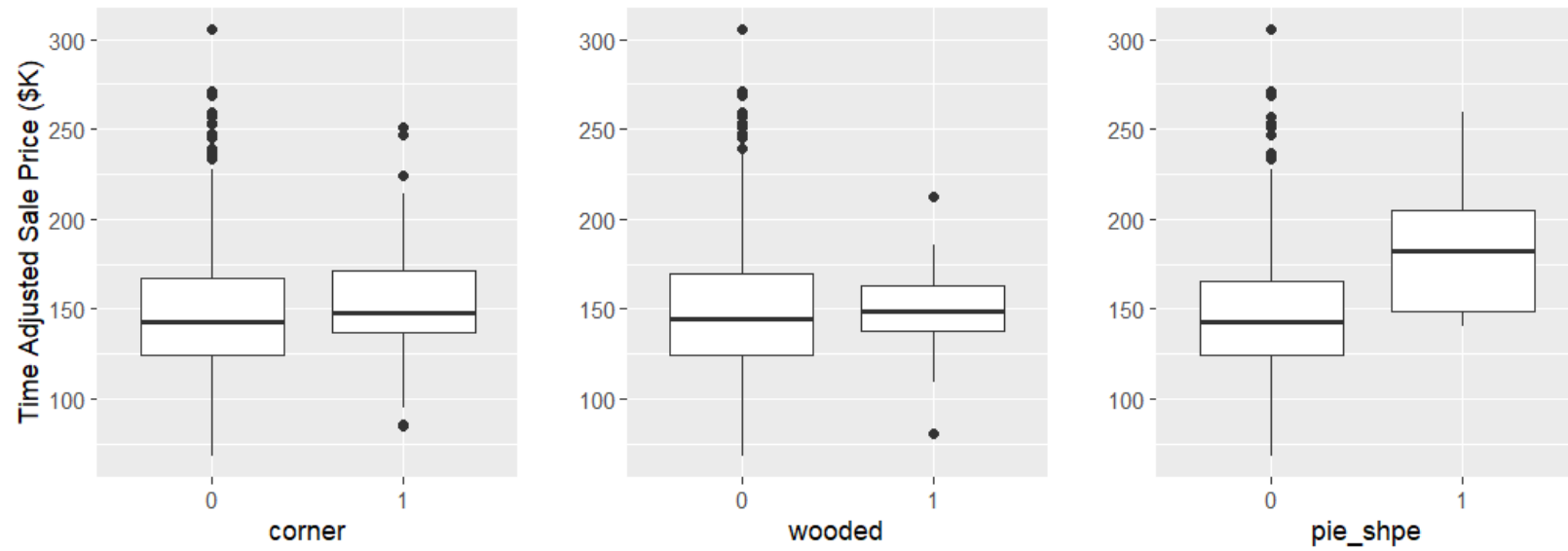


Fig A 4. Variable Correlation with Response Variable – Low Correlation



A.3. Step 7 – Model Calibration

Stepwise Regression

Table A 2 – Model Summary for the Models between Steps

Model	R	R Squared	Adjusted R Squared	Std. Error of the Estimate	F Statistic
Model1	0.897	0.805	0.797	16564.210	102.295
Model2	0.897	0.805	0.797	16545.990	108.550
Model3	0.897	0.804	0.797	16551.060	115.184
Model4	0.896	0.804	0.797	16557.110	122.684
Model5	0.896	0.802	0.796	16601.870	130.495

Table A 3 Model Variables for the Models between Steps

Model	Predictors:
Model1	familyrm, flrarea2_adjusted, lot_size_adjusted, fireplcs, baseste, pie_shpe, heattype, effage, culdesac, splitlvl, sidewalk, garagefl, area1_2_adjusted, bsmtfin_adjusted, crwlarea, conduit, bedrooms, baths
Model2	familyrm, flrarea2_adjusted, lot_size_adjusted, fireplcs, baseste, pie_shpe, effage, culdesac, splitlvl, sidewalk, garagefl, area1_2_adjusted, bsmtfin_adjusted, crwlarea, conduit, bedrooms, baths
Model3	familyrm, lot_size_adjusted, fireplcs, baseste, pie_shpe, effage, culdesac, splitlvl, sidewalk, garagefl, area1_2_adjusted, bsmtfin_adjusted, crwlarea, conduit, bedrooms, baths
Model4	familyrm, lot_size_adjusted, fireplcs, baseste, pie_shpe, effage, splitlvl, sidewalk, garagefl, area1_2_adjusted, bsmtfin_adjusted, crwlarea, conduit, bedrooms, baths
Model5	familyrm, lot_size_adjusted, fireplcs, baseste, pie_shpe, effage, splitlvl, sidewalk, garagefl, area1_2_adjusted, bsmtfin_adjusted, crwlarea, conduit, baths

A.4. Step 8 – Model Testing

Model Adjustment

Table A 4. *Ratio Statistics for ASR after Adjustment*

Neigh	Median	Lower Bound	Upper Bound	Actual Coverage	Minimum	Maximum	COD
713	0.99	0.96	1.03	95.6	0.78	1.32	8.56
715	1	0.97	1.05	94.6	0.8	1.28	7.45
716	1	0.99	1.02	95.3	0.82	1.22	5.95
731	1	0.94	1.05	92.2	0.9	1.18	7.59
734	1	0.92	1.03	93.5	0.85	1.16	7.16
736	1.09	0.97	1.1	93.6	0.89	1.24	6.65
737	0.99	0.96	1.01	93.7	0.76	1.22	7.94

Table A 5. *Ratio Statistics for ASR before Manual Class Adjustment*

Manual Class	Median	Lower Bound	Upper Bound	Minimum	Maximum	COD
40	0.92	0.85	0.93	0.85	0.93	1.95
51	0.85	0.85	0.85	0.85	0.85	0
80	1.02	0.98	1.09	0.91	1.37	8.69
81	0.94	0.78	1.13	0.78	1.13	7.58
90	1	1	1.02	0.76	1.24	7.05
92	1.05	0.97	1.28	0.97	1.28	7.18
140	1	0.98	1.01	0.79	1.22	6.73
142	0.91	0.86	0.97	0.86	0.97	4.44
145	0.93	0.82	1.03	0.82	1.03	6.08

Table A 6. Ratio Statistics for ASR after Manual Class Adjustment

Manual Class	Median	Lower Bound	Upper Bound	Minimum	Maximum	COD
40	1.01	0.92	1.01	0.92	1.01	1.95
51	1	1	1	1	1	0
80	0.98	0.94	1.05	0.87	1.32	8.69
81	0.94	0.78	1.13	0.78	1.13	7.58
90	1	1	1.02	0.76	1.24	7.05
92	1.05	0.97	1.28	0.97	1.28	7.18
140	1	0.98	1.01	0.79	1.22	6.73
142	1	0.95	1.07	0.95	1.07	4.44
145	0.93	0.82	1.03	0.82	1.03	6.08

Table A 7. No Adjustment for heatttype, bedroom, culdesac, and conduit

	heatttype	bedroom	culdesac	alley	conduit
Kruskal-Wallis H	4.52	5.58	0.14	0.19	0.60
df	5	2	1	1	1
Asymp Sig.	0.48	0.07	0.71	0.66	0.43

Table A 8. Ratio Statistics for ASR before sidewalk Adjustment

Sidewalk	Median	Lower Bound	Upper Bound	Minimum	Maximum	COD
0	1	0.99	1.01	0.72	1.4	7.41
1	1.06	1	1.1	0.85	1.34	8.02

Table A 9. Ratio Statistics for ASR after sidewalk Adjustment

Sidewalk	Median	Lower Bound	Upper Bound	Minimum	Maximum	COD
0	1	0.99	1.01	0.72	1.4	7.41
1	1	0.95	1.04	0.8	1.26	8.02

Syntax (in R)

```
library(tidyverse)
library(readr)
library(haven)
library(magrittr)
library(car)
library(patchwork)
library(corrplot)
library(rlang)
library(broom)
rm(list = ls())
```

```
data <- read_sav("Data/BUSI 444 Term 2 Case Study 466 Marilu Project 2.sav")
```

```
data%>%mutate(SAR = saleprice/totalval)
```

```
SAR_month <- data%>%group_by(month)%>%summarise(SAR = mean(SAR), .groups = 'drop')
```

```
SAR_July_1<-(SAR_month[SAR_month['month']==6,'SAR'] +
SAR_month[SAR_month['month']==7,'SAR'])$SAR/2
```

```
SAR_month%>%mutate(Factor = SAR_July_1 / SAR)
```

```
data%>%mutate(
  saleprice_time_adjusted = case_when(
    month == 3 ~ saleprice * 1.013,
    month == 4 ~ saleprice * 0.999,
    month == 5 ~ saleprice * 1.001,
    month == 6 ~ saleprice * 1.009,
    month == 7 ~ saleprice * 0.991,
    month == 8 ~ saleprice * 1.011,
```

```

    month == 9 ~ saleprice * 0.977,
    month == 10 ~ saleprice * 0.979
  )
)

kruskal.test(data$saleprice_time_adjusted, data$month)

View(SAR_month)

n<-nrow(data)
lower_index <- qbinom(0.05 / 2, n, 0.5)
upper_index <- qbinom(1- 0.05/2, n, 0.5)

data%<>%mutate(landsize_per_square = landval/lot_size)
data%<>% mutate(sizefact_eq = (mean(lot_size)/lot_size)^(1.2/2))
data%<>% mutate(sizefact = landsize_per_square/median(landsize_per_square))

data%>%ggplot(aes(x = lot_size/10000, y = sizefact))+ geom_point()+
  geom_line(aes(x=lot_size/10000, y= sizefact_eq), size = 1, color = 'red')+
  labs(x = 'Lot area in sq feet (divided by 10,000)', y= 'Size Factor',
       title = 'Size Factor Vs. Lot Area')

# epsilon_values <- seq(0.5, 2, by=0.1) # Try values between 0.5 and 2
# errors <- numeric(length(epsilon_values))
#
# for (i in seq_along(epsilon_values)) {
#   e <- epsilon_values[i]

```

```

# adjusted_size <- data$sizefact_eq^(e/2) # Apply exponent
# errors[i] <- sum((adjusted_size - data$sizefact)^2) # Replace expected_values
with real data
# }

# best_epsilon <- epsilon_values[which.min(errors)]
# print(best_epsilon)

## Fitting Quadratic, Cubic, and Power models to the data
power_model <- lm(log(sizefact) ~ log(lot_size/10000), data = data)

a <- exp(coef(power_model)[1]) # Intercept
b <- coef(power_model)[2]      # Slope
data <- data %>%
  mutate(adjs_fac = a * ((lot_size/10000)^b),
         lot_size_adjusted = lot_size * adjs_fac)

## Manual Classes
data %<>%
  mutate(
    linmcls_cp = case_when(
      cp_mancl %in% c(910,911) ~ 0.450,
      cp_mancl %in% c(920,921) ~ 1.000,
      cp_mancl %in% c(930,931) ~ 1.200,
      cp_mancl %in% c(940,941) ~ 1.420,
      TRUE ~ 1.000
    ),
    linmcls_gr = case_when(
      gr_mancl %in% c(910,911) ~ 0.450,
      gr_mancl %in% c(920,921) ~ 1.000,
      gr_mancl %in% c(930,931) ~ 1.200,

```

```

    gr_manc1 %in% c(940,941) ~ 1.420,
    TRUE ~ 1.000
  ))

```

```

rank_no <- function (data, x) {
  x <- enquo(x)
  output <- rbind(
    data%>%mutate(Rank = rank(!x), value = corner)%>%group_by(value)%>%
      summarise(count = n(), MeanRank = mean(Rank)),
    data%>%mutate(Rank = rank(!x), value = culdesac)%>%group_by(value)%>%
      summarise(count = n(), MeanRank = mean(Rank)),
    data%>%mutate(Rank = rank(!x), value = alley)%>%group_by(value)%>%
      summarise(count = n(), MeanRank = mean(Rank)),
    data%>%mutate(Rank = rank(!x), value = wooded)%>%group_by(value)%>%
      summarise(count = n(), MeanRank = mean(Rank)),
    data%>%mutate(Rank = rank(!x), value = pie_shpe)%>%group_by(value)%>%
      summarise(count = n(), MeanRank = mean(Rank)),
    data%>%mutate(Rank = rank(!x), value = slope)%>%group_by(value)%>%
      summarise(count = n(), MeanRank = mean(Rank)),
    data%>%mutate(Rank = rank(!x), value = abv_road)%>%group_by(value)%>%
      summarise(count = n(), MeanRank = mean(Rank)))
  return(output)
}

```

```

ratio_statistics <- function (data, x, y) {
  x <- enquo(x)
  y <- enquo(y)

  output <- round(cbind(data%>%
    arrange(!x)%>%

```

```

group_by(!y, .drop = FALSE)%>%reframe(
  Mean = mean(!x),
  LB_mean = mean(!x) - 1.96 * sd(!x)/sqrt(n()),
  UB_mean = mean(!x) + 1.96 * sd(!x)/sqrt(n()),
  Median = median(!x),
  Min = min(!x),
  Max = max(!x),
  COD = (mean(abs(!x-median(!x)))/median(!x)) * 100,
  CV = sd(!x)/mean(!x)*100,
  LB_mode = (!x)[max(qbinom(0.05 / 2, n(), 0.5),1)],
  UB_mode = (!x)[max(qbinom(1-0.05 / 2, n(), 0.5),1)],
  Actual_coverage = (pbinom(max(qbinom(1-0.05 / 2, n(),
0.5),1), n(), 0.5) -
                                                                    pbinom(max(qbinom(0.05 / 2, n(),
0.5),1), n(), 0.5))*100)),2)
  return (output)
}

```

EDA

```

data%<>%mutate(flarea1_adjusted = linmcls * flarea1,
  flarea2_adjusted = linmcls * flarea2,
  area1_2_adjusted = linmcls * area1_2,
  totarea_adjusted = linmcls * totarea,
  bsmtfin_adjusted = linmcls * bsmtfin,
  gar_area_adjusted = linmcls_gr * gar_area,
  cp_area_adjusted = linmcls_cp * cp_area,
  effage = 2017 - mbeffyr)%>%
  mutate(neigh_713 = ifelse(neigh == 713, 1,0),
    neigh_715 = ifelse(neigh == 715, 1,0),
    neigh_731 = ifelse(neigh == 731, 1,0),

```

```

    neigh_732 = ifelse(neigh == 732, 1,0),
    neigh_734 = ifelse(neigh == 734, 1,0),
    neigh_736 = ifelse(neigh == 736, 1,0),
    neigh_737 = ifelse(neigh == 737, 1,0))

data_model <- data%>%select(bedrooms, baths, familyrm, fireplcs,
    flrarea1_adjusted,flrarea2_adjusted,
                                bsmtfin_adjusted, crwlarea, bsmtarea, unfinbmt,
    area1_2_adjusted,totarea_adjusted, baseste,
                                heattype, splitlvl,lot_size_adjusted,
                                corner, wooded, pie_shpe, slope, abv_road,
    culdesac,effage,
                                conduit, alley, sidewalk, gar_area_adjusted, garagefl,
    cp_area_adjusted, carptfl,
                                saleprice_time_adjusted, neigh_713, neigh_715, neigh_731,
    neigh_732, neigh_734, neigh_736, neigh_737)

View(cor(data_model))

data%>%group_by(neigh)%>%summarise(N = n(),
                                Mean = mean(saleprice_time_adjusted),
                                std.Deviation = sd(saleprice_time_adjusted),
                                .groups = 'drop')

scatter_plots <- function (data, x, y_label){
  x <- enquo(x)
  fig <- data%>%ggplot(aes(x= !!x, y= saleprice_time_adjusted/1000))+geom_point()+
    geom_smooth(method = 'lm', se = FALSE)

  if (y_label) {
    fig <- fig + labs(y = "Time Adjusted Sale Price ($K)")
  }
}

```

```

} else {
  fig <- fig + labs(y = "")
}

formula <- new_formula(
  lhs = quote(saleprice_time_adjusted),
  rhs = get_expr(x),
  env = caller_env()
)

r_squared <- summary(lm(formula, data = data))$r.squared
return (list(figure = fig, r_sq = r_squared))
}

box_plots <- function (data, x, y_label){
  x <- enquo(x)

  data %<>%mutate(!!x := as.factor(!!x))

  fig <- data%>%
    ggplot(aes(x= !!x, y= saleprice_time_adjusted/1000))+geom_boxplot()

  if (y_label) {
    fig <- fig + labs(y = "Time Adjusted Sale Price ($K)")
  } else {
    fig <- fig + labs(y = "")
  }

  return (list(figure = fig))
}

```

```
}
```

```
### Descriptive Analysis
```

```
descriptive_analyzis <- sapply(X= names(data_model), FUN = function (x) {
```

```
  return(list(
    Min = min(data_model[[x]], na.rm = TRUE),
    Max = max(data_model[[x]], na.rm = TRUE),
    Mean = mean(data_model[[x]], na.rm = TRUE),
    Median = median (data_model[[x]], na.rm = TRUE)
```

```
  ))
```

```
})
```

```
View(as.data.frame(descriptive_analyzis)%>%t%>%as.data.frame()%>%filter(Max > 10))
```

```
Descriptive_cols <-
```

```
as.data.frame(descriptive_analyzis)%>%t%>%as.data.frame()%>%filter(Max <= 10)
```

```
frequency_table <- lapply(row.names(Descriptive_cols), function(x) {
```

```
  # Summarise the count for each group in the column
```

```
  result <- data_model %>%
```

```
    group_by(!sym(x)) %>%
```

```
    summarise(count = n(), .groups = 'drop')
```

```
  } )
```

```
## Fixing heatttype
```

```
data%>%mutate(heatttype = case_when(  
  heatttype == 1 ~ 2,  
  heatttype == 5 ~ 6,  
  TRUE ~ heatttype  
)
```

```
## Effective Age
```

```
effective_year_plot <- scatter_plots(data, effage, TRUE)$figure  
effective_year_r_sq <- scatter_plots(data, effage, TRUE)$r_sq
```

```
alley_plot <- box_plots(data, alley, TRUE)$figure
```

```
## Crawl Area
```

```
crawl__plot <- scatter_plots(data%>%filter(crawlarea>0), crawlarea, TRUE)$figure  
crawl_r_sq <- scatter_plots(data%>%filter(crawlarea>0), crawlarea, TRUE)$r_sq
```

```
## Side Walk
```

```
Side_walk <- box_plots(data, sidewalk, FALSE)$figure
```

```
## Basement Presence
```

```
Bastement_presence <- box_plots(data, baseste, FALSE)$figure
```

```
## Above Road
```

```
Above_rd <- box_plots(data, abv_road, TRUE)$figure
```

```
## Slope
```

```
Slope <- box_plots(data, slope, FALSE)$figure
```

```
## Wooden
```

```
Wooded <- box_plots(data, wooded, FALSE)$figure
```

```
## Corner
```

```
Corner <- box_plots(data, corner, TRUE)$figure
```

```
## Heattype
```

```
Heat_type <- box_plots(data, heattype, TRUE)$figure
```

```
## Carport Presence
```

```
Carport_presence <- box_plots(data, carptfl, FALSE)$figure
```

```
## Crawl Area
```

```
Crawl_area__plot <- scatter_plots(data%>%filter(crawlarea>0), crawlarea, FALSE)$figure
```

```
Crawl_area_r_sq <- scatter_plots(data%>%filter(crawlarea>0), crawlarea, TRUE)$r_sq
```

```
## Carport Area
```

```
Carport_area__plot <- scatter_plots(data%>%filter(cp_area_adjusted>0),  
cp_area_adjusted, FALSE)$figure
```

```
Carport_area_r_sq <- scatter_plots(data%>%filter(cp_area_adjusted>0),  
cp_area_adjusted, TRUE)$r_sq
```

```
## Unfinished Basement
```

```
unfinished_area_plot <- scatter_plots(data%>%filter(unfinbmt>0), unfinbmt,  
FALSE)$figure
```

```
unfinished_area_r_sq <- scatter_plots(data%>%filter(unfinbmt>0), unfinbmt, TRUE)$r_sq
```

```
## Pie_shape
```

```
Pie_shape <- box_plots(data, pie_shpe, FALSE)$figure
```

```
## Garage Area
```

```
gar_area_plot <- scatter_plots(data%>%filter(gar_area_adjusted>0), gar_area_adjusted,  
TRUE)$figure
```

```
gar_area_r_sq <- scatter_plots(data%>%filter(gar_area_adjusted>0), gar_area_adjusted,  
TRUE)$r_sq
```

```
## Floor Area 2
```

```
floor_2_plot <- scatter_plots(data%>%filter(flarea2_adjusted>0), flarea2_adjusted,  
FALSE)$figure
```

```
floor_2_r_sq <- scatter_plots(data%>%filter(flarea2_adjusted>0), flarea2_adjusted,  
TRUE)$r_sq
```

```
## Garage Presence
```

```
Garage_presence <- box_plots(data, garagefl, FALSE)$figure
```

```
## Split-level Presence
```

```
Split_presence <- box_plots(data, splitlvl, FALSE)$figure
```

```
## Culdesac Presence
```

```
Culdesac_presence <- box_plots(data, culdesac, TRUE)$figure
```

```
## Bedrooms
```

```
Bedrooms <- box_plots(data, bedrooms, FALSE)$figure
```

```
## Lot Size
```

```
lot_plot <- scatter_plots(data%>%filter(lot_size_adjusted>0), lot_size_adjusted,  
TRUE)$figure
```

```
lot_r_sq <- scatter_plots(data%>%filter(lot_size_adjusted>0), lot_size_adjusted,  
TRUE)$r_sq
```

```
## FamilyRoom
```

```
Family_room <- box_plots(data, familyrm, FALSE)$figure
```

```
## Basement Finished
```

```
Basement_plot <- scatter_plots(data%>%filter(bsmtfin_adjusted>0), bsmtfin_adjusted,  
FALSE)$figure
```

```
Basement_r_sq <- scatter_plots(data%>%filter(bsmtfin_adjusted>0), bsmtfin_adjusted,  
FALSE)$r_sq
```

```
## Fireplcs
```

```
Fireplcs <- box_plots(data, fireplcs, TRUE)$figure
```

```
## Conduits
```

```
Conduits <- box_plots(data, conduit, FALSE)$figure
```

```
## Basement Area
```

```
Basement_area_plot <- scatter_plots(data%>%filter(bsmtarea>0), bsmtarea,  
FALSE)$figure
```

```
Basement_area_r_sq <- scatter_plots(data%>%filter(bsmtarea>0), bsmtarea, TRUE)$r_sq
```

```
## Baths
```

```
Bath_plot <- box_plots(data, baths , TRUE)$figure
```

```
## Total Area
```

```
Total_area_plot <- scatter_plots(data%>%filter(totarea_adjusted>0), totarea_adjusted,  
TRUE)$figure
```

```
Total_area_r_sq <- scatter_plots(data%>%filter(totarea_adjusted>0), totarea_adjusted,  
TRUE)$r_sq
```

```
## Area_1_2
```

```
Area_1_2_plot <- scatter_plots(data%>%filter(area1_2_adjusted>0), area1_2_adjusted,  
FALSE)$figure
```

```
Area_1_2_r_sq <- scatter_plots(data%>%filter(area1_2_adjusted>0), area1_2_adjusted,  
TRUE)$r_sq
```

```
## Floor Area_1
```

```
Floor_1_plot <- scatter_plots(data%>%filter(flarea1_adjusted>0), flarea1_adjusted,  
FALSE)$figure
```

```
Floor_1_r_sq <- scatter_plots(data%>%filter(flarea1_adjusted>0), flarea1_adjusted,  
TRUE)$r_sq
```

```
## Green
```

```
(Total_area_plot | Area_1_2_plot | Floor_1_plot) /  
  (Bath_plot | Basement_area_plot | effective_year_plot)
```

```
Total_area_r_sq
```

```
Area_1_2_r_sq
```

```
Floor_1_r_sq
```

```
Basement_area_r_sq
```

```
effective_year_r_sq
```

```
box_plots(data, neigh, TRUE)$figure
```

```
## Yellow
```

```
(Culdesac_presence | Bedrooms | Family_room)/  
(Fireplcs | Conduits |Pie_shape)
```

```
(lot_plot | Basement_plot | unfinished_area_plot)/  
  (gar_area_plot | floor_2_plot | Garage_presence)
```

```
lot_r_sq  
Basement_r_sq  
unfinished_area_r_sq  
gar_area_r_sq  
floor_2_r_sq
```

```
Basement_r_sq  
crawl_r_sq
```

```
Corner | Wooded | Pie_shape  
## White
```

```
(Heat_type | Carport_presence | Crawl_area__plot)/  
(alley_plot | Split_presence |Bastement_presence)/  
  (Above_rd | Slope | Side_walk)
```

```
Carport_area_r_sq  
Crawl_area_r_sq
```

```
## Data Model
```

```

data_model <- data%>%select(bedrooms, baths, familyrm, fireplcs,
flrarea1_adjusted,flrarea2_adjusted,
                        bsmtfin_adjusted, crwlnarea, bsmtarea, unfinbmt,
area1_2_adjusted,totarea_adjusted, baseste,
                        heattype, splitlvl,lot_size_adjusted,
                        corner, wooded, pie_shpe, slope, abv_road,
culdesac,effage,
                        conduit, alley, sidewalk, gar_area_adjusted, garagefl,
cp_area_adjusted, carptfl,
                        neigh,
                        saleprice_time_adjusted)

data_group_1 <- data_model%>% select(familyrm, flrarea2_adjusted, lot_size_adjusted,
                        fireplcs, baseste, pie_shpe,
                        heattype, effage, culdesac,
                        splitlvl, sidewalk, garagefl,
                        saleprice_time_adjusted,
                        flrarea1_adjusted, bsmtfin_adjusted, crwlnarea,
alley,
                        bedrooms, baths)

model_group_1 <- lm(saleprice_time_adjusted ~ ., data = data_group_1)

data_group_2 <- data_model%>% select(familyrm, flrarea2_adjusted, lot_size_adjusted,
                        fireplcs, baseste, pie_shpe,
                        heattype, effage, culdesac,
                        splitlvl, sidewalk, garagefl,
                        saleprice_time_adjusted,
                        bsmtarea, bsmtfin_adjusted, conduit,
                        bedrooms, baths)

model_group_2 <- lm(saleprice_time_adjusted ~ ., data = data_group_2)

```

```

data_group_3 <- data_model%>% select(familyrm, flrarea2_adjusted, lot_size_adjusted,
                                     fireplcs, baseste, pie_shpe,
                                     heattype, effage, culdesac,
                                     splitlvl, sidewalk, garagefl,
                                     saleprice_time_adjusted,
                                     totarea_adjusted,    crwllarea,    conduit,
                                     bedrooms, baths)
model_group_3 <- lm(saleprice_time_adjusted ~ ., data = data_group_3)

```

```

data_group_4 <- data_model%>% select(familyrm, flrarea2_adjusted, lot_size_adjusted,
                                     fireplcs, baseste, pie_shpe,
                                     heattype, effage, culdesac,
                                     splitlvl, sidewalk, garagefl,
                                     saleprice_time_adjusted,
                                     area1_2_adjusted, bsmtfin_adjusted , crwllarea
                                     ,conduit,
                                     bedrooms, baths)

```

```

model_group_4 <- lm(saleprice_time_adjusted ~ ., data = data_group_4)

```

```

model_names <- paste0("model_group_", 1:4)

```

```

adjusted_r2 <- c()
mean_sse <- c()
f_stat <- c()
p_value <- c()

# Loop through each model and extract metrics
for (i in 1:4) {
  # Get the model dynamically
  model <- get(model_names[i])
  model_summary <- summary(model)

  # Extract residual sum of squares (SSE)
  sse <- sum(model_summary$residuals^2)

  # Residual degrees of freedom (df)
  df_residual <- model_summary$df[2]

  # Calculate Mean SSE (MSE)
  mean_sse[i] <- sse / df_residual

  # Extract other metrics
  adjusted_r2[i] <- model_summary$adj.r.squared
  f_stat[i] <- model_summary$fstatistic[1]
  p_value[i] <- pf(model_summary$fstatistic[1],
                    model_summary$fstatistic[2],
                    model_summary$fstatistic[3],
                    lower.tail = FALSE)
}

# Create the final data frame

```

```

results_table <- data.frame(
  Group = model_names,
  Adjusted_R2 = adjusted_r2,
  Mean_SSE = mean_sse,
  F_Statistic = f_stat,
  Significance = p_value
)

print(results_table)

model <- model_group_4
model_summary <- summary(model)

# Extract coefficients
coefficients_table <- as.data.frame(model_summary$coefficients)

# Create base table
result_table <- data.frame(
  Variable = rownames(coefficients_table),
  Coefficient = coefficients_table$Estimate,
  Standard_Coefficient = coefficients_table$Estimate / sd(model$model[[1]]), #
  Standardized Coefficient
  t_Statistic = coefficients_table$`t value`,
  Significance = coefficients_table$`Pr(>|t|)`
)

# Calculate VIF and Tolerance
vif_values <- vif(model)
tolerance <- 1 / vif_values

```

```

# Add VIF and Tolerance to the table
result_table$Tolerance <- c(NA, tolerance) # NA for intercept
result_table$VIF <- c(NA, vif_values)      # NA for intercept

# View the result table
View(result_table)

## MODEL/TEST Split

data_model <- cbind(data_group_4, random = data$random, neigh = data$neigh)

data_train <- data_model%>%arrange(random)%>%filter(row_number() <= 320)%>%select(-
random)
data_test <- data_model%>%arrange(random)%>%filter(row_number() > 320)%>%select(-
random)

## Model

# Stepwise regression using AIC for model selection
full_model <- lm(saleprice_time_adjusted ~ ., data = data_train%>%select(-neigh))
null_model <- lm(saleprice_time_adjusted ~ 1, data = data_train%>%select(-neigh))

# Initialize a list to store intermediate models
intermediate_models <- list()

# Custom function to capture intermediate models

```

```

capture_models <- function(model, direction) {
  intermediate_models <- append(intermediate_models, list(model))
}

# Perform stepwise regression with tracing
step_model <- step(full_model, direction = "both", trace = 1, scope = list(lower =
null_model, upper = full_model), k = 2,
                    steps = 1000, keep = capture_models)

# Extract formulas and refit models
model_formulas <- lapply(intermediate_models, formula)
models <- lapply(model_formulas, function(formula) lm(formula, data = data))

# Assign names to the models
names(models) <- paste0("Model", seq_along(models))

model_stats_step <- bind_rows(lapply(models, glance)) %>%
  mutate(Model = names(models)) %>%
  select(Model, r.squared, adj.r.squared, sigma, statistic) %>%
  rename(
    `R Squared` = r.squared,
    `Adjusted R Squared` = adj.r.squared,
    `Std. Error of the Estimate` = sigma,
    `F Statistic` = statistic
  ) %>%
  mutate(R = sqrt(`R Squared`)) %>%
  select(Model, R, `R Squared`, `Adjusted R Squared`, `Std. Error of the Estimate`,
`F Statistic`)

View(model_stats_step)

```

```

coefficients_steps <- model_stats_step %>%
  mutate(Variables = sapply(models, function(model) {
    terms <- labels(terms(model)) # Extract variable names
    paste(terms, collapse = ", ") # Combine into a single string
  })) %>%
  select(Model, Variables)

View(coefficients_steps)

## Final Model
final_model <- intermediate_models[[length(intermediate_models)]]

# Get ANOVA table for the final model
anova_table <- anova(final_model)
SSE_total = anova_table$`Sum Sq`
Df_totak = anova_table$Df

data.frame(SSE = c(sum(SSE_total[1:length(SSE_total)-
1]),SSE_total[length(SSE_total)], sum(SSE_total)),
           df = c(sum(Df_totak[1:length(Df_totak)-1]),Df_totak[length(Df_totak)],
sum(Df_totak)),
           F_value = c(mean(anova_table$`F value`, na.rm = TRUE), NULL, NULL))

# Get Coefficient

model <- final_model
model_summary <- summary(model)

# Extract coefficients
coefficients_table <- as.data.frame(model_summary$coefficients)

```

```

# Create base table
result_table <- data.frame(
  Variable = rownames(coefficients_table),
  Coefficient = coefficients_table$Estimate,
  Standard_Coefficient = coefficients_table$Estimate / sd(model$model[[1]]), #
  Standardized Coefficient
  t_Statistic = coefficients_table$t value`,
  Significance = coefficients_table$`Pr(>|t|)`
)

vif_values <- vif(model)
tolerance <- 1 / vif_values

# Add VIF and Tolerance to the table
result_table$Tolerance <- c(NA, tolerance) # NA for intercept
result_table$VIF <- c(NA, vif_values)      # NA for intercept

# View the result table
View(result_table)

# Casewise Diagnosis
model <- final_model

# Calculate standardized residuals
std_residuals <- rstandard(model)

# Get the response variable (y) and predicted values
response_variable <- model$model$saleprice_time_adjusted # Actual response variable
predicted_values <- fitted(model) # Predicted values from the model

```

```

residuals <- residuals(model)          # Residuals (actual - predicted)

# Create a data frame with casewise diagnostics
diagnostics_df <- data.frame(
  Case = 1:length(std_residuals), # Case number
  Std_Residuals = std_residuals,  # Standardized residuals
  Response_Variable = response_variable, # Actual response variable
  Predicted_Values = predicted_values,  # Predicted values
  Residuals = residuals             # Residuals
)

# Filter cases with standardized residuals > 3 or < -3
diagnosed_cases <- diagnostics_df %>%
  filter(Std_Residuals > 3 | Std_Residuals < -3)

# Display the diagnosed cases
print(diagnosed_cases)

# Residuals
predicted_values <- fitted(model) # Predicted values
residuals <- residuals(model)     # Residuals
std_predicted <- scale(predicted_values) # Standardized predicted values
std_residuals <- rstandard(model) # Standardized residuals

# Create a data frame with the required metrics
metrics_table <- data.frame(
  Metric = c("Predicted Values", "Residuals", "Std. Predicted Values", "Std.
Residuals"),
  Min = c(min(predicted_values), min(residuals), min(std_predicted),
min(std_residuals)),

```

```

    Max = c(max(predicted_values), max(residuals), max(std_predicted),
max(std_residuals)),

    Mean = c(mean(predicted_values), mean(residuals), mean(std_predicted),
mean(std_residuals)),

    Std_Deviation = c(sd(predicted_values), sd(residuals), sd(std_predicted),
sd(std_residuals)),

    Count = c(length(predicted_values), length(residuals), length(std_predicted),
length(std_residuals))
)

# Rename columns for better readability
colnames(metrics_table) <- c("Metric", "Min", "Max", "Mean", "Std. Deviation",
"Count")

# Display the table
print(metrics_table)

model_summary <- summary(model)
r_value <- sqrt(model_summary$r.squared)
r_squared <- model_summary$r.squared
adj_r_squared <- model_summary$adj.r.squared
std_error <- model_summary$sigma

metrics_table_2 <- data.frame(
  Metric = c("R", "R-squared", "Adjusted R-squared", "Std. Error of the Estimate"),
  Value = c(r_value, r_squared, adj_r_squared, std_error)
)

# Display both tables
print("Table 1: Predicted Values, Residuals, Std. Predicted, Std. Residuals")
print(t(metrics_table_2))

```

```

data_train <- data%>%arrange(random)%>%filter(row_number() <= 320)

data_train$Prediction <- predict(model, newdata = data_train)
data_train%<>%mutate(ASR = Prediction/saleprice_time_adjusted)

data_test$Prediction <- predict(model, newdata = data_test)
data_test%<>%mutate(ASR = Prediction/saleprice_time_adjusted)

data$Prediction <- predict(model, newdata = data)
data%<>%mutate(ASR = Prediction/saleprice_time_adjusted)

data_train%>%mutate(Rank = rank(ASR), value = neigh)%>%group_by(value)%>%
  summarise(count = n(), MeanRank = mean(Rank))

## Neighborhood Adjustment
kruskal.test(data_train$ASR, data_train$neigh)

ratio_statistics(data_train, ASR, neigh)

data_train%<>%mutate(Prediction_neigh = case_when(
  neigh == 715 ~ Prediction / 1.08,
  neigh == 731 ~ Prediction / 0.83,
  neigh == 734 ~ Prediction / 1.31,
  TRUE ~ Prediction),
  ASR_neigh = Prediction_neigh/saleprice_time_adjusted)

data%<>%mutate(Prediction_neigh = case_when(
  neigh == 715 ~ Prediction / 1.08,
  neigh == 731 ~ Prediction / 0.83,

```

```

neigh == 734 ~ Prediction / 1.31,
TRUE ~ Prediction),
ASR_neigh = Prediction_neigh/saleprice_time_adjusted)

kruskal.test(data_train$ASR_neigh, data_train$neigh)
ratio_statistics(data_train, ASR_neigh, neigh)

## Manual Class Adjustment
kruskal.test(data_train$ASR_neigh, data_train$mbmancls)

ratio_statistics(data_train, ASR_neigh, mbmancls)

data_train%<>%mutate(Prediction_neigh_mclass = case_when(
  mbmancls == 40 ~ Prediction_neigh / 0.92 ,
  mbmancls == 51 ~ Prediction_neigh / 0.85 ,
  mbmancls == 142 ~ Prediction_neigh / 0.91 ,
  TRUE ~ Prediction_neigh),
ASR_neigh_mclass = Prediction_neigh_mclass/saleprice_time_adjusted)

data%<>%mutate(Prediction_neigh_mclass = case_when(
  mbmancls == 40 ~ Prediction_neigh / 0.92 ,
  mbmancls == 51 ~ Prediction_neigh / 0.85 ,
  mbmancls == 142 ~ Prediction_neigh / 0.91 ,
  TRUE ~ Prediction_neigh),
ASR_neigh_mclass = Prediction_neigh_mclass/saleprice_time_adjusted)

kruskal.test(data_train$ASR_neigh_mclass, data_train$mbmancls)
ratio_statistics(data_train, ASR_neigh_mclass, mbmancls)

```

```
## Bedroom
```

```
kruskal.test(data_train$ASR_neigh_mclass, data_train$bedrooms)
```

```
ratio_statistics(data_train, ASR_neigh_mclass, bedrooms)
```

```
## Heatttype
```

```
kruskal.test(data_train$ASR_neigh_mclass, data_train$heatttype)
```

```
ratio_statistics(data_train, ASR_neigh_mclass, heatttype)
```

```
## Culdesac
```

```
kruskal.test(data_train$ASR_neigh_mclass, data_train$culdesac)
```

```
ratio_statistics(data_train, ASR_neigh_mclass, culdesac)
```

```
## Alley
```

```
kruskal.test(data_train$ASR_neigh_mclass, data_train$alley)
```

```
ratio_statistics(data_train, ASR_neigh_mclass, alley)
```

```
## Entire Dataset
```

```
ratio_statistics(data, ASR_neigh_mclass)
```

```
## Modeling ASR
```

```
data_asr <- data%>%select(colnames(data_group_4), ASR_neigh_mclass)%>%  
  select(-saleprice_time_adjusted)
```

```
asr_model <- lm(ASR_neigh_mclass ~ ., data = data_asr)
```

```

model <- asr_model
model_summary <- summary(model)

# Extract coefficients
coefficients_table <- as.data.frame(model_summary$coefficients)

# Create base table
result_table <- data.frame(
  Variable = rownames(coefficients_table),
  Coefficient = coefficients_table$Estimate,
  Standard_Coefficient = coefficients_table$Estimate / sd(model$model[[1]]), #
  Standardized Coefficient
  t_Statistic = coefficients_table$t value`,
  Significance = coefficients_table$`Pr(>|t|)`
)

# Calculate VIF and Tolerance
vif_values <- vif(model)
tolerance <- 1 / vif_values

# Add VIF and Tolerance to the table
result_table$Tolerance <- c(NA, tolerance) # NA for intercept
result_table$VIF <- c(NA, vif_values)      # NA for intercept

# View the result table
View(result_table)

## conduit

```

```

kruskal.test(data$ASR_neigh_mclass, data$conduit)
ratio_statistics(data, ASR_neigh_mclass, conduit)

## sidewalk

kruskal.test(data$ASR_neigh_mclass, data$sidewalk)
ratio_statistics(data, ASR_neigh_mclass, sidewalk)

data%<>%mutate(Prediction_neigh_mclass_sidewalk = case_when(
  sidewalk == 1 ~ Prediction_neigh_mclass / 1.06 ,
  TRUE ~ Prediction_neigh_mclass),
  ASR_neigh_mclass_sidewalk =
Prediction_neigh_mclass_sidewalk/saleprice_time_adjusted)

kruskal.test(data$ASR_neigh_mclass_sidewalk, data$sidewalk)
ratio_statistics(data, ASR_neigh_mclass_sidewalk, sidewalk)

## effage and flrarea2_adjusted

effage_plot_2 <- data%>%ggplot(aes(x = effage, y = ASR_neigh_mclass_sidewalk)) +
  geom_point()+ geom_smooth(method = "lm", se = FALSE)+
  labs(y = "Adjusted ASR")

summary(lm(ASR_neigh_mclass_sidewalk ~ effage, data = data))$r.squared

flrarea2_plot<- data%>%ggplot(aes(x = flrarea2_adjusted, y =
ASR_neigh_mclass_sidewalk)) +
  geom_point()+ geom_smooth(method = "lm", se = FALSE)+
  labs(y = "Adjusted ASR")

```

```
summary(lm(ASR_neigh_mclass_sidewalk ~ flrarea2_adjusted, data = data))$r.squared
```

```
effage_plot_2 | flrarea2_plot
```

```
### Final check with neighborhood
```

```
kruskal.test(data_train$ASR_neigh_mclass, data_train$neigh)
```